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Edited by:

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Message from the Chairs

On behalf of the entire organizing team of ESEC/FSE 2019 it is a great honor and pleasure for us to welcome you to the 27th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE) held on 26-30 August 2019 in Tallinn, Estonia.

The conference brings together researchers, practitioners, and educators in software engineering to discuss the most recent innovations and challenges in the area. This year's program continues the great tradition of previous ESEC/FSE joint meetings in bringing together and exchanging the latest research results and trends, as well as their practical application in all areas of software engineering.

The research track attracted 303 submissions from all over the world. The reviewing process was performed by a program committee composed of 95 of the world's top software engineering experts from academia and industry. This year, we followed a full double-blind review process (author identities were not disclosed at any point during the process). Each submission received reviews from, at least, three members of the program committee, as well as one meta-review explaining the final decision. Authors of the submitted papers were allowed to provide a response to the reviews during a rebuttal phase, followed by an extensive online discussion of the papers by the respective reviewers and additional discussants from the program committee. The program committee eventually accepted 74 papers covering a large variety of software engineering topics, including empirical studies, program analysis, testing, modeling, search-based software engineering, software architecture, software product lines, and many more. Finally, out of the 74 accepted papers, six were selected to achieve ACM SIGSOFT Distinguished Paper Awards.

This year's ESEC/FSE again encouraged authors of accepted papers to submit data sets and any additional artefacts needed to replicate their work to the artifact evaluation track. This year, the track implements ACM's "Result and Artifact Review and Badging" policy, which recognizes high quality artifacts through official badges. The "Artifacts Evaluated - Functional" badge indicates that an artifact is complete, well documented and allows to obtain the research track paper's results, while the "Artifacts Evaluated - Reusable" badge indicates that an artifact is of such a high quality that it can be reused as is on other datasets, or for other purposes. Of course, every artifact had to be easy to use or analyze, and should be accessible from a public archival repository for at least 5 years. Any research track paper with an accepted artifact is being highlighted in the conference program.

Out of the 74 accepted research track papers (each eligible to submit to the artifacts track), 21 artifact submissions were received, ranging from data sets to benchmarks or tools. Each artifact was reviewed by one artifact PC member (out of a committee of 10 international experts), after which the authors were able to react to these reviews by fixing their artifacts or their documentation. Subsequent re-reviewing and discussion led to 19 artifacts being accepted, 17 of which obtaining the available badge, while 15 were of such high quality that they obtained a reusable badge. Given that the whole reviewing procedure (including rebuttal phase) took place in a short time span, we would like to thank the authors and reviewers for their dedication! Finally, we extend a special word of thanks to the

authors of the reusable badge artifacts for their extra effort in advancing the state-of-the-art in the community.

ESEC/FSE 2019 continues to support the Journal First Initiative initiated some years ago in the SIGSOFT community. With this initiative, selected journal papers that are not extensions of conference papers are presented at one of the leading conferences in the area. This allows attendees to hear talks about excellent research work from world-leading journals that has only previously appeared in a journal and would, therefore, otherwise go without being presented at any conference. This year, the conference invited the authors of all eligible papers of the ACM Transactions on Software Engineering and Methodology (TOSEM), the IEEE Transactions on Software Engineering (TSE), and the Springer Journal on Empirical Software Engineering (EMSE) to present. In addition, we invited peer reviewed papers published in IEEE Software to be presented as part of the industry track. In total, we received 27 proposals of Journal First papers, from which we invited 24 to be presented at the conference.

The conference also attracted papers with strong focus on real-world applicability of software engineering techniques, tools, methodologies, processes, and practices. In total, 21 industry papers were selected out of 39 submissions. In addition, contributions to a tool demo track were attracted resulting in a total of 15 demonstrations, accompanied by separate presentations, selected out of 33 submissions. As in previous conferences, we organized a student research competition with the participation of 29 candidate students (selected out of 39 submissions), and a doctoral symposium track, held on the day before the main conference, in which ten research proposals (selected out of 13 submissions) were discussed and detailed feedback provided to the students. In addition, this year, the doctoral symposium features a keynote talk given by Brian Fitzgerald, Director of Lero, the Irish Research Software Research Centre, titled *Advice on Conducting Software Engineering Research (and Getting it Published).*

The main conference features altogether six invited keynote talks: two keynote talks from the research track entitled *Living with Feature Interactions* by Joanne Atlee (University of Waterloo) and *Safety and Robustness for Deep Learning with Provable Guarantees* by Marta Kwiatkowska (University of Oxford); one keynote talk from the ACM SIGSOFT Impact Paper Award, entitled *CUTE: A Concolic Unit Testing Engine for C* by Koushik Sen, Darko Marinov and Gul Agha; two keynote talks from the ESEC/FSE Test of Time Award, entitled *Cross-project Defect Prediction: a Large Scale Experiment on Data vs. Domain vs. Process* by Thomas Zimmermann, Nachiappan Nagappan, Harald Gall, Emanuel Giger, and Brendan Murphy, and *Fair and balanced?: Bias in Bug-fix Datasets* by Christian Bird, Adrian Bachmann, Eirik Aune, John Duffy, Abraham Bernstein, Vladimir Filkov, and Premkumar Devanbu; and one keynote talk from the industry track, entitled *Insights from Open Source Software Supply Chains* by Audris Mockus (University of Tennessee).

Besides the traditional doctoral symposium, the pre-conference activities include a record number of seven workshops on a set of important topics: the 10th Int. Workshop on Automating Test Case

Design, Selection and Evaluation (A-Test), the 2nd Int. Workshop on Software-intensive Business: Start-ups, Platforms and Ecosystems (IWSiB), 1st Int. Workshop on Education through Advanced Software Engineering and Artificial Intelligence (EASEAI), 2nd Int. Workshop on Ensemble-based Software Engineering for Modern Computing Platforms (EnSEmble), 2nd Int. Workshop on Software Qualities and their Dependencies (SQUADE), 3rd Int. Machine Learning Techniques for Software Quality Evaluation (MaLTeSQuE), and the 3rd Int. Workshop on App Market Analytics (WAMA). The conference is also co-located with the Annual Symposium on Search-Based Software Engineering (SSBSE), to be held right after ESEC/FSE.

We are extremely grateful to the various people who made this great event possible, and we apologize in advance for not mentioning everyone by name. We are extremely thankful to the members of the organizing committee who worked hard to the development of all parts of the conference and we extend a special thanks to the program committee members of all tracks who worked hard to select the most inspiring contributions which formed the basis for putting together the final program.

We would like to acknowledge ACM for the sponsorship of ESEC/FSE 2019 through its Special Interest Group on Software Engineering (SIGSOFT). We also deeply thank our supporters Microsoft, Google, Huawei, IBM Research, and ING. With this sponsorship, financial support, and many contributions from members of the software engineering community, we are happy to present to you an exciting and rich program with ideas that we are sure will have great impact and will foster progress in software engineering worldwide.

Finally, we want to thank all of the conference attendees for contributing to making ESEC/FSE 2019 a success. We hope that you will find the program interesting, thought-provoking, and inspiring, and that the conference will give you valuable opportunities to share ideas with researchers and practitioners from the vibrant software engineering community.

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